

**ADDIS ABABA UNIVERSITY,
FACULTY OF SCIENCE**



Department of Mathematics and Computer Science

**Entrance examination for the Masters of Science
Programme in Computer Science**

Date: April 28, 2003
Time allowed: 3 hours

Full Name : _____

Grade Obtained : _____

Important Note

- **Write your name only on this page. Writing your name on the other pages will automatically disqualify you from this exam.**
- Answer neatly to all the questions on the space provided on the question paper
- Write short, and grammatically and logically correct sentences

Part I: Computer organization and assembly (15 points total) **(29 Minutes Total)**

a) **(4 points)** Give two applications where writing the code in assembly language rather than in high-level language is compulsory or highly beneficial. Explain very succinctly why it is compulsory or highly beneficial.

(4 Minutes)

#	Application	Reason
1		
2		

b) **(2 points)** Give two major concepts that make the difference between RISC and CISC technologies and how these concepts contribute in improving the performance of the processor. **(4 Minutes)**

c) **(1 point)** Give the name of an existing processor that is build based on RISC technology. **(1 Minutes)**

d) **(2 points)** Give two major functions of the CPU. **(2 Minutes)**

#	Application
1	

e) (2 Points) Give the **name** of the character encoding system that is most widely used for encoding of texts in most alphabets and scripts of the World. What are the **differences** between this system and the Latin character encoding system almost exclusively used, until recently, to encode texts in Latin characters? What are its **advantages** and its **drawbacks**? Answer very succinctly in the table. (3 Minutes)

Name of the character encoding system	
Differences	
Advantages of the character encoding system	
Drawbacks of the character encoding system	

f) (4 points) (15 Minutes) We would like to write a fragment of code of the virtual memory manager of an operating system in INTEL 8086 assembly language. The objective is to write a fragment of the assembly language code that calculates the physical (real) address corresponding to a given virtual address. Write this fragment of code knowing that:

- The virtual address is in AX
- The page size is 256 Bytes
- The page table is in memory at the address PAGE_TABLE. Each entry of the page table occupies one byte and contains only the page frame number.
- We consider that all pages of the virtual memory are always in the real memory (there is no page fault)
- The result (real address) should be in BX
- You should use the simplest and minimum number of instructions in order to minimize the execution time of this code
- If there is a mnemonic of an instruction that you have forgotten use INT1, INST2, etc. and explain what this instruction does. This will be accepted only if an instruction that does exactly what you say exists in INTEL 8086.

Explain how your program works and give your commented program.

Explanation

Program

Part II: Programming (15 points total)

Part III: Software Engineering (15 points total)

Part IV: Database Systems (14 points total)

Part V: Operating system (11 points total) (23 Minutes Total)

- 1) **(3 points) (5 Minutes)** Input/outputs in general and disk in particular are the major bottlenecks of the computer system. Give two very efficient mechanisms used by the operating system to limit this problem. Explain briefly how each of the mechanism contributes for this purpose.

- 2) **(6 points) (15 Minutes)** Write a program in C programming language that runs on UNIX or Linux systems. The program requests 10 whole numbers from the user, finds the maximum of the ten numbers and the sum and displays both. The programs should have three processes. The first reads the input from the user. The second finds the maximum and displays it. The third calculates the sum and displays it.

Note

- For the communication and synchronization between the processes you can use:
 - i. semaphores (you can use any commonly syntax for semaphores not necessarily found on UNIX/Linux)
 - ii. signals
 - iii. pipes
 - iv. parent/child inheritance
- Proper comments are necessary.

- 3) **(2 points) (3 Minutes)** Many client/server applications such as web servers create one thread to process the request of a client at the server side than creating a process for each client. Explain briefly why?

Part VI: Networking (15 points total)

Part VII: General questions on ICT in Ethiopia and abroad (15 points total) (15 Minutes Total)

Instruction: Attempt any two of the following five questions. **If you attempt more than two none will be accepted.**

Each question is given 7.5 points as long as you follow the above instructions.

1. Some years back, the Federal Government and many of the state of the United States of America sued Microsoft. What was the reason why they sued this company? Microsoft seems to have escaped the worst. Do you think the World has anything to lose or gain from this trial's result. Argument your answer in about 10 lines (average handwriting).

2. What is E-commerce? Do you think it has a great future in the World in general and in Ethiopia in particular. What are the technical issues that have to be considered if we want to see the development of E-commerce in Ethiopia. Argument your answer in about 10 lines (average handwriting).